

Bob, Verified.

A harness that turns Bob's code review into something you can check — not a paragraph of AI prose you're asked to take on faith.

hash-chained audit trail

deterministic risk classification

filler-stripped output

THE PROBLEM

AI agents are graded on what they say happened — not what actually did.

Nothing forces an agent's summary to line up with the real state of the filesystem, and nothing stops that summary from padding a thin result with confident-sounding filler.

For a code review tool specifically, that gap is the whole product failing at its one job: you end up trusting the reviewer's word about the review.

WHAT THIS HARNESS ADDS

Bob's Agent mode, wrapped by two zero-dependency verification libraries



sipa-trace

Every tool call becomes a hash-chained TraceCard, risk-classified by capability (read/write/delete/network/credentials) — never by Bob grading its own action. `verify()` confirms nothing was edited after the fact.

sipa-signal

Bob's own written explanation is run through a deterministic pattern-match pass, stripping hedging and filler before a human reads it. Not another model summarizing Bob's summary.

WHY THE CLASSIFIER IS GENERIC

Classifies the action, never the task.

risk_map.py looks at what KIND of action a tool call is — read, write, delete, network, credentials — never the domain of the task Bob was given. The same classifier works whether Bob is reviewing a payments service or a static blog. Unknown tools fall through to NONE risk by design: an unrecognised action gets flagged for a human, never waved through as either safe or dangerous.

TOOL PATTERN	EXAMPLE	RISK CLASS
read-only	read_file	NONE
writes / edits	write_file, apply_patch	LOW
network	web_fetch, curl	MEDIUM
deletes / destructive shell	rm -rf, git reset --hard	CRITICAL
touches credentials	any param mentioning token/key/secret	escalated

REAL RUN, NOT A MOCKUP

Verified live, 2026-09-26.



Bob reviewed a sample calc.py. 1 tool call (read_file), correctly classified NONE. Hash chain verified clean. Review text passed through sipa-signal: classified CLEAN, 0% filler detected in this run.

```
task_id: 6b97a4360ede8290bbf159f8d7cd0fde
cost:    0.144 Bobcoins (real spend)
verify:  VerifyResult(ok=True, count=1, problems=[])
signal:  CLEAN, 0% filler
```

TRY IT

Demo & source

LIVE DEMO

bobsprint.sipa-os.org

REPOSITORY

github.com/soulinpsyabstract/sipa-bob-sprint

Built for the IBM Bob 2.0 Hackathon (lablab.ai)

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